
AI for Ecosystem Simulation and Optimization

Huzaifa Iqbal (31563)

Huzaifa.Iqbal.31563@khi.iba.edu.pk

Muhammad Usman Hussain (28992)

m.hussain.28992@khi.iba.edu.pk

Ali Hamza (30618)

a.hamza.30618@khi.iba.edu.pk

Daniyaal Bokhari (31571)

Daniyaal.bokhari.31571@khi.iba.edu.pk

CSE-307 Introduction to Artificial Intelligence

IBA Karachi

May 6, 2026

Abstract

Wildlife conservation faces a fundamental challenge that is managing the reintroduction of species into fragile habitats without causing irreversible ecological collapse. Ecologists refer to this as the Yellowstone Problem. When predators such as wolves are reintroduced into a ecosystem, even a small miscalculation in starting population numbers can be catastrophic. Too many predators quickly exhaust the prey population, and the few, unchecked herbivores strip the land bare. Both outcomes end in total ecosystem failure. Current conservation strategies rely heavily on historical precedent and expert judgment, making them both slow and risky to deploy in the field.

This project addresses the problem directly by developing an AI-Optimized Ecosystem Simulation designed to serve as an Ecological Decision Support System. We built a dynamic, grid-based simulation engine that models the interactions between plants, herbivores, and carnivores across 120 simulated months, governed by 24 changeable environmental parameters. To find the ideal balance of these parameters, we implemented four AI algorithms entirely from scratch in Python, without relying on any external machine learning libraries: a Genetic Algorithm (GA) and Particle Swarm Optimization (PSO) for parameter optimization, a feedforward Neural Network (NN) for predictive forecasting, and a Decision Tree (DT) for extracting actionable conservation rules.

Our key findings show that the GA consistently outperformed PSO in maximizing a composite fitness score that weighs survival duration, biodiversity, and population stability. The Decision Tree extracted clear, human-readable policies from simulation data, with one particularly important finding: introducing carnivores with high initial energy reserves into small, constrained habitats results in a 100% ecosystem collapse rate. Ultimately, this project demonstrates that a unified AI system can discover, predict, and explain the optimal conditions needed to sustain long-term ecological balance, replacing dangerous trial-and-error conservation with fast, safe, computational testing.

Keywords: Ecosystem Simulation, Genetic Algorithm, Particle Swarm Optimization, Neural Networks, Decision Trees, Conservation AI

1 Introduction

1.1 Background and Motivation

Ecosystems are among the most complex systems in nature. Every organism, from the smallest blade of grass to the apex predator at the top of the food chain, depends on a delicate and interconnected balance to survive. A single change, whether a prolonged drought, the removal of a keystone species, or the arrival of a new predator, can ripple through the entire food web and transform a thriving habitat into a barren wasteland within just a few seasons.

Historically, ecologists have relied on field observations, long-term population surveys, and expert intuition to understand these dynamics. While invaluable, these methods have serious limitations. They are slow, expensive, and ultimately reactive so we often only understand what went wrong after the damage is done. As climate change accelerates habitat loss and human expansion fragments natural reserves, the urgency of developing faster, more reliable tools for ecological decision-making has never been greater.

This project was motivated by one central question which was “Can Artificial Intelligence replace the guesswork in wildlife conservation?”. By building a digital ecosystem where thousands of scenarios can be tested safely in software, we can give conservationists a powerful tool to evaluate strategies before deploying them in the real world, where mistakes cost millions of dollars and result in the loss of irreplaceable animal life.

1.2 Problem Statement and Objectives

The central problem this project addresses is known among ecologists as the Yellowstone Problem. The name comes from the real-world reintroduction of wolves into Yellowstone National Park, a landmark conservation effort that required years of careful planning precisely because the stakes of getting the numbers wrong were so high. In general, any time a conservationist tries to reintroduce a species into a habitat, they face a critical question that “How many animals, at what starting health, energy levels and on what size of land? The answer must be nearly exact. If too many predators are introduced, they quickly hunt herbivores to extinction and then starve. If too few, herbivores overpopulate and destroy all the vegetation, leaving no food for anyone.

Trial-and-error in real ecosystems is not an option. Each failed experiment costs millions in funding, destroys years of conservation progress, and causes the needless death of animals. Our primary objective was to eliminate this dangerous guesswork.

To achieve this, we set the following specific goals:

- Build a realistic, dynamic grid-based ecosystem simulation that models plant growth, herbivore grazing, and carnivore hunting.
- Implement a Genetic Algorithm (GA) and Particle Swarm Optimization (PSO) from scratch to search across 24 environmental parameters and find the configurations that maximize long-term ecological health.

-
- Train a Neural Network (NN) on simulation data so that it can instantly predict whether a given habitat configuration will survive without needing to run a full simulation.
 - Build a Decision Tree (DT) that analyses simulation outcomes and automatically extracts simple and human-readable conservation policies that a park ranger can act on directly.
 - Compare the performance of GA and PSO and evaluate the accuracy and interpretability of the NN and DT models.

2 Related Work

The application of Artificial Intelligence to ecological problems has grown substantially in recent years. Researchers across multiple disciplines have turned to computational methods to simulate, predict, and manage biological systems that are too complex or dangerous to study purely through fieldwork. The three areas most directly relevant to this project are evolutionary computation for ecological optimization, machine learning for wildlife prediction, and decision-support frameworks for conservation policy.

2.1 Evolutionary Algorithms in Predator-Prey Dynamics

One of the most direct precursors to our approach is the work by Talas et al. [1], who adapted Genetic Algorithms to simulate natural selection in a real predator-prey context. In their study, wild birds acted as the selective pressure as they were allowed to hunt artificial prey whose visual patterns were controlled by a GA. The survival or capture of each artificial prey item determined which visual traits were passed on to the next generation. Over many iterations, the GA evolved prey camouflage patterns that genuinely deceived real predators.

This work demonstrates two things that are directly relevant to our project. First, it validates that GAs are capable of evolving solutions to complex biological problems when given a biologically meaningful fitness signal. Second, it shows that the GA framework can interface with real-world selection pressure, not just mathematical functions. Our project extends this principle from the micro-level (evolving the visual pattern of individual prey) to the macro-level. Rather than evolving a single trait of a single organism, our GA evolves the overarching environmental and population parameters of an entire ecosystem by starting populations, energy reserves, reproduction thresholds, and seasonal growth rates. The fitness signal is not survival of individual prey items, but the long-term health and biodiversity of the entire simulated habitat.

2.2 Predictive Machine Learning in Wildlife Conservation

A second major strand of relevant research concerns the use of machine learning, and Neural Networks in particular, to make sense of ecological data. Tuia et al. [2] published a comprehensive perspective paper in *Nature Communications* arguing that machine learning is fundamentally transforming wildlife conservation. They highlight how modern sensor technologies, camera traps, acoustic monitors, and satellite imagery generate far more data than human analysts can process. Neural Networks, they argue, are uniquely suited to finding non-linear patterns in this high-dimensional data, enabling applications from automated species identification to habitat connectivity mapping and population trend forecasting.

This paper directly motivates the Neural Network component of our project. Rather than processing sensor data from real animals, however, our NN is trained on hundreds of outcomes generated by our own simulator. Each training example consists of 24 ecosystem parameters as input and the number of months the ecosystem survived as output. Once trained, the NN can instantly predict the survival duration of any new parameter configuration, replacing a computationally expensive simulation with a near-instant estimate. This makes the NN an

essential practical tool because it acts as a fast model that conservationists can use to screen large numbers of habitat configurations before committing to a full simulation run.

2.3 Decision Trees for Conservation Policy

The third relevant area of prior work concerns how to translate complex analytical findings into clear, actionable policies for people working in the field. Tulloch et al. [3] addressed this challenge directly in the context of biodiversity data publication. They developed a Decision Tree framework to help conservationists decide whether publishing the GPS locations of endangered species would help or harm those species, weighing factors such as the risk of poaching against the benefits of increased public awareness and scientific collaboration.

What makes their work particularly instructive is their choice of tool. A Decision Tree was selected not because it is the most accurate possible classifier, but because its output, a branching series of IF/THEN rules, can be understood and applied by non-specialists without any data science background. A park ranger in the field does not have access to a laptop running a neural network so they need a laminated card with a clear decision flowchart.

Our project integrates this principle directly. After our GA and PSO identify high-performing ecosystem configurations and our simulator generates hundreds of labelled outcomes, our Decision Tree analyses this data and automatically extracts the most important rules governing ecosystem survival. The result is a set of concrete, plain-language policies, for example, specifying the maximum number of carnivores a given grid size can support before the ecosystem collapses. This ensures our AI system does not remain a black box, but produces outputs that conservation practitioners can immediately understand and use.

2.4 Positioning of This Work

While each of the above papers applies AI to one aspect of ecological management, our project combines all three approaches into a single, unified decision support system. The GA and PSO handle optimization, the Neural Network handles prediction, and the Decision Tree handles explanation and policy extraction. Together, they form an end-to-end pipeline that not only finds the best ecosystem configuration but also tells the user why it works and what to avoid. This integration is the primary contribution of our project.

3 Methodology

The methodology of this project has two major components. The first is the ecosystem simulation engine: the digital environment in which all experiments take place. The second is the AI pipeline which is the four algorithms that interact with the simulator to find, predict, and explain optimal conservation strategies. A core constraint for this project is that every component, the simulation engine and all four AI algorithms, was written entirely from scratch in Python using only NumPy for numerical computation. No machine learning libraries were used at any point.

3.1 The Ecosystem Simulation Engine

The simulation environment is a two-dimensional grid representing a physical wildlife reserve. The grid can be configured from a minimum of 10 x 10 cells up to a maximum of 40 x 40 cells, giving a total reserve area of between 100 and 1,600 habitat units. Each cell can hold any number of organisms simultaneously, and the simulation advances in discrete monthly time steps for a total of 120 simulated months, representing a ten-year observation period.

Three types of entities inhabit the grid. Plants are stationary organisms that grow, spread to neighbouring cells based on the current season, and die of old age or from winter die-off events. Herbivores are mobile animals that move around the grid searching for plants to eat, consume energy as they move, reproduce when their energy exceeds a threshold, and die of old age, starvation, or predation. Carnivores are also mobile and follow a similar lifecycle, but instead of eating plants, they hunt herbivores. Every month, the simulation updates all three populations in sequence that is plants grow and spread, herbivores graze and move, and carnivores hunt and reproduce.

The entire behaviour of this system is controlled by 24 numerical parameters, which the AI algorithms manipulate in order to find configurations that lead to stable, long-lived ecosystems. These parameters are described in detail in Section 3.2.

The Fitness Formula

To give the AI algorithms a single, clear objective to optimize, we designed a composite fitness score that rates each simulation outcome on a scale from 0.0 to 1.0. This score is calculated from three components. The first and most heavily weighted component, accounting for 50% of the total score, is survival duration which tells how many of the 120 simulated months the ecosystem remained alive without any species going completely extinct. The second component, weighted at 30%, is the Shannon Diversity Index, a standard ecological metric that rewards configurations where all three species maintain healthy population sizes, penalizing outcomes where one species dominates while others barely survive. The third component, weighted at 20%, is population stability which is a penalty for wild boom-and-bust cycles where populations swing dramatically from one extreme to another each month. A stable ecosystem earns a higher score even if its populations are smaller, because stability is a sign of genuine long-term balance rather than temporary survival.

3.2 Critical System Parameters

The simulation is governed by 24 parameters in total. Rather than describing all of them at once, this section focuses on five critical parameters meaning those whose values have the largest single impact on ecosystem health and whose correct setting is most difficult to determine without computational assistance.

Carnivore Start Energy (carn_start_energy, range: 75–135)

This parameter sets the amount of energy each carnivore has at the moment, it is introduced into the ecosystem. It is arguably the single most consequential parameter in the entire system. A carnivore with very high starting energy can hunt aggressively for many months before needing to worry about starvation. In a small habitat with a limited herbivore population, this means a high-energy predator will exhaust the prey supply long before the herbivore population has had a chance to recover through reproduction. The ecosystem collapses from the top down. Conversely, if carnivores start with very low energy, they may starve before successfully hunting enough prey to sustain themselves, leading to predator extinction, herbivore overpopulation, and eventual plant collapse from the bottom up. The safe window for this parameter is therefore quite narrow, and it is strongly dependent on the grid size, the initial herbivore count, and the hunt success rate. Our Decision Tree analysis revealed that a carnivore start energy above approximately 120 units in a grid smaller than 15 x 15 cells results in a 100% ecosystem collapse rate.

Hunt Success Rate (hunt_rate, range: 0.38–0.92)

This parameter controls the probability that a carnivore successfully catches and kills a herbivore during a hunting attempt. It acts as a direct multiplier on predator effectiveness. A high hunt success rate means carnivores are extremely efficient predators, depleting herbivore populations rapidly. A low rate means carnivores spend energy moving and attempting hunts without catching enough food, eventually starving. The parameter is critical because it mediates the entire predator-prey balance. Ecosystems with an intermediate hunt success rate, roughly 0.55 to 0.70 in our experiments, showed the highest long-term stability, because prey populations had enough breathing room to reproduce and recover between hunting events. Very high values consistently led to herbivore extinction while very low values led to carnivore extinction.

Initial Population Counts (initial_herbivores, initial_carnivores)

The starting number of herbivores and carnivores represents the most direct expression of the Yellowstone Problem in our simulation. These two parameters interact closely. The ratio of predators to prey at the start of the simulation is often more important than the absolute number of either. In our experiments, we found that introducing even two or three additional carnivores beyond the ideal ratio could tip a stable ecosystem into collapse within the first twenty simulated months. The initial plant count is also important, but plants are more resilient because they reproduce passively based on season. Herbivores and carnivores, by contrast, require careful calibration against each other and against the grid size.

Herbivore Reproduction Threshold (herb_repro_thresh, range: 55–135)

A herbivore reproduces when its current energy level exceeds this threshold. This parameter controls how quickly the prey population can grow and recover after a period of heavy

predation. A low threshold means herbivores reproduce frequently, which can sustain a large carnivore population but also risks herbivore overpopulation if predator numbers drop. A high threshold means herbivores reproduce rarely, keeping the population lean, but making it fragile while a brief period of intense predation can crash the herbivore count below the recovery threshold, leading to extinction. Finding the right value requires matching this parameter to the hunt success rate and the carnivore start energy, which is precisely the kind of multi-variable optimization that the GA and PSO are designed to handle.

Plant Seasonal Spread Rates (plant_spread_spring through plant_spread_winter)

Plants do not grow at a constant rate in our simulation. They spread to neighbouring cells at different rates depending on the current season like fastest in spring and summer, slower in autumn, and at a near-halt in winter. Together, these four spread rate parameters define the rhythm of the entire ecosystem. If spring and summer spread rates are high, the plant population rebounds quickly after winter, giving herbivores a reliable food supply. If winter die-off rates are also high, the ecosystem experiences sharp seasonal population swings that can cascade into predator starvation. The seasonal parameters matter because they create the cyclical pressure that all other species must adapt to. An ecosystem configured with parameters calibrated only for average conditions, without accounting for seasonal extremes, will often fail during its first simulated winter.

3.3 Optimization Algorithms

Genetic Algorithm (GA)

The Genetic Algorithm takes its inspiration from biological evolution. The core idea is simple that if you repeatedly select the best-performing individuals from a population, let them reproduce with each other, and introduce occasional random variation, the population will gradually evolve toward better and better solutions.

In our implementation, a population consists of 40 randomly generated sets of ecosystem parameters. Each parameter set is evaluated by running a complete 120-month simulation and recording the resulting fitness score. The parameter sets with the highest fitness scores are treated as parents for the next generation. We use Tournament Selection to pick parents, and we randomly draw three candidates from the population and select the one with the highest fitness score. This process is repeated to select two parents.

The two parents then combine their parameters through Uniform Crossover. For each of the 24 parameters, a coin is flipped to decide which parent passes that parameter to each child. This produces two offspring that are each a random mix of their two parents. Finally, each offspring undergoes Gaussian Mutation with a 10% probability per parameter, and the parameter value is shifted by a small random amount drawn from a normal distribution centred at zero, introducing novel variation that helps the algorithm avoid getting stuck in local optima.

One final mechanism (Elitism) ensures that the single best parameter set from each generation is always carried into the next generation unchanged. This prevents the algorithm from accidentally losing its best solution through crossover or mutation. Over 30 generations, this

evolutionary process consistently converges on parameter configurations that achieve high fitness scores.

The GA was chosen for this problem because the parameter search space is extremely large (24 continuous and integer-valued parameters with complex non-linear interactions), and evolutionary algorithms are known to perform well in such high-dimensional spaces. They do not require the fitness landscape to be smooth or differentiable, which is important because our fitness function is produced by a stochastic simulation with many interacting components.

Particle Swarm Optimization (PSO)

Particle Swarm Optimization is a different approach to the same problem. Instead of biology, it takes its metaphor from the flocking behaviour of birds or the schooling of fish. The key observation is that a flock of birds searching for food tends to converge on the best food source because every bird shares information with its neighbours about what it has found.

In our implementation, the population consists of 30 particles, each representing a random set of ecosystem parameters. Every particle has both a position (its current parameter values) and a velocity (the direction and speed at which its parameters are changing). At each iteration, every particle evaluates its current position by running a simulation, then updates its velocity based on two forces. The first force, called the cognitive component, pulls the particle toward the best position it has personally found so far. The second force, called the social component, pulls it toward the best position found by any particle in the entire swarm. A third factor, the inertia weight, controls how much of the particle's previous velocity it retains. We linearly decrease the inertia weight from 0.9 to 0.4 over the course of the optimization, allowing the swarm to explore broadly at the start and converge more precisely toward the end.

PSO was chosen as a comparison algorithm to the GA because it has a fundamentally different search dynamic. While the GA maintains diversity through crossover and mutation, PSO converges through collaborative information sharing. In our experiments, PSO converged somewhat faster in the early iterations but tended to plateau at a lower final fitness score than the GA, suggesting that the ecosystem parameter space benefits from the GA's mutation-driven ability to escape local optima.

Artificial Neural Network (NN)

Running a full 120-month simulation for every parameter configuration that needs to be evaluated is computationally expensive. The Neural Network addresses this by learning to predict the outcome of a simulation instantly, without actually running it.

All 24 input parameters are standardized to zero mean and unit variance before training to ensure that parameters with large numerical ranges do not dominate those with small ones. We train the network on a dataset of several hundred simulation outcomes collected from the GA and PSO runs, using 80% for training and 20% for validation.

Once trained, the NN can evaluate a new parameter configuration in milliseconds instead of the several seconds required for a full simulation. This makes it practical as a first-pass screening tool which a conservationist can use the NN to quickly filter thousands of possible configurations, then run full simulations only on the most promising ones.

Decision Tree (DT)

While the Neural Network provides fast and accurate predictions, it cannot explain its reasoning. A conservationist cannot use it to understand why a configuration succeeds or fails, only whether it will. The Decision Tree was added specifically to solve this problem.

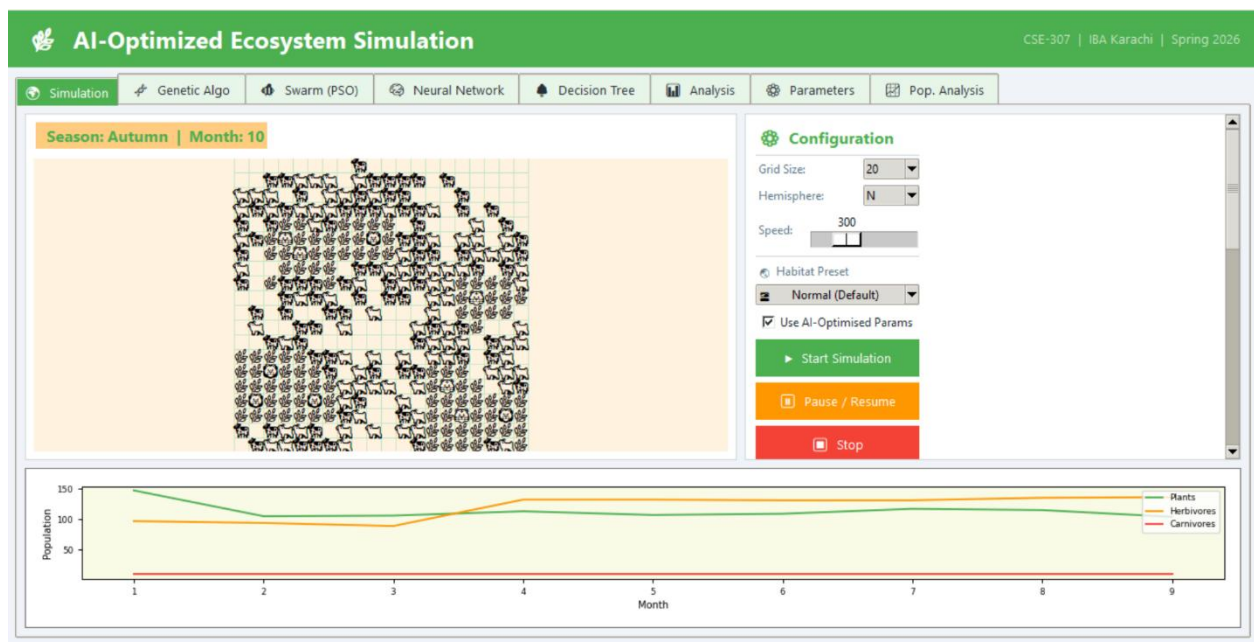
It classifies each configuration as either Survived (more than 60 months) or Collapsed (60 months or fewer). The tree is built recursively meaning at each node, it searches through all 24 parameters and selects the one whose threshold split best separates the Survived outcomes from the Collapsed outcomes. This is measured using Information Gain, a quantity derived from the entropy of the outcome distribution. A split that perfectly separates the two classes produces the maximum information gain.

The result is a tree of IF/THEN decision rules. For example, IF carnivore start energy > 120 AND grid size < 15, THEN the ecosystem will collapse. These rules can be read and applied by anyone, regardless of their background in machine learning. They represent our system's most direct and practical output for real-world conservation use.

4 Results and Discussion

4.1 Simulation Results Overview

Running the full ecosystem simulation pipeline across both optimization algorithms, multiple grid sizes, and both hemispheres produced a rich dataset of outcomes that clearly shows the value of AI-guided parameter search. In the base configuration (a 20×20 grid, Northern Hemisphere, default parameters), unoptimized random parameter sets frequently collapsed within the first 20 to 40 simulated months, and rarely kept all three species alive beyond month 60. Once the GA or PSO applied their optimized parameter sets, survival durations improved dramatically. The best GA-optimized configurations achieved a perfect composite fitness score of **1.0000** and kept the full simulation running for all 120 months.



The figure illustrates what a successfully running ecosystem looks like. The grid is densely populated with herbivores and plants, and the population timeline below shows all three species maintaining stable, overlapping trajectories throughout the first nine months. Plants (green line) remain the most abundant layer, herbivores (orange) settle at a healthy intermediate level, and carnivores (red) hold a low but consistent presence. This three-tier population structure with many plants, moderate herbivores, and few but stable carnivores is the hallmark of a well-calibrated ecosystem and the target that all four AI components were designed to achieve.



The scalability experiment revealed an important finding about larger reserves consistently enabled longer survival and higher fitness scores. As shown in the Scalability Analysis, survival duration increased from 38 months on a 10×10 grid (100 km²) to 55 months on a 40×40 grid (1,600 km²). Fitness scores rose accordingly from 0.336 to 0.422. Smaller grids pack interactions so tightly that a brief surge in predator efficiency can wipe out prey before they have room to spread and recover. Larger grids give populations the space to escape local hunting pressure, which translates directly into greater resilience. This finding has a clear practical implication that reserve size is itself a conservation tool, and our simulation puts a number on the survival benefit of each incremental increase in protected area.

4.2 Neural Network Results

The Neural Network was trained on 80 labelled simulation outcomes (80% training, 20% validation) over 60 epochs at a learning rate of 0.001. Its job was to predict how many months a given 24-parameter configuration would sustain the ecosystem without needing a full simulation run.



The Neural Network panel presents three key diagnostics. The **Training Loss curve** (top left) shows training loss (solid purple) and validation loss (dashed blue) both declining steadily from around 2.5 down to below 1.0 and 0.8 respectively over 60 epochs. The smooth, parallel descent of these two curves confirms that the network learned genuine patterns rather than memorizing the training data. This is a result of meaningful signal in the 24-parameter feature space combined with adequate regularization from the standardization preprocessing step.

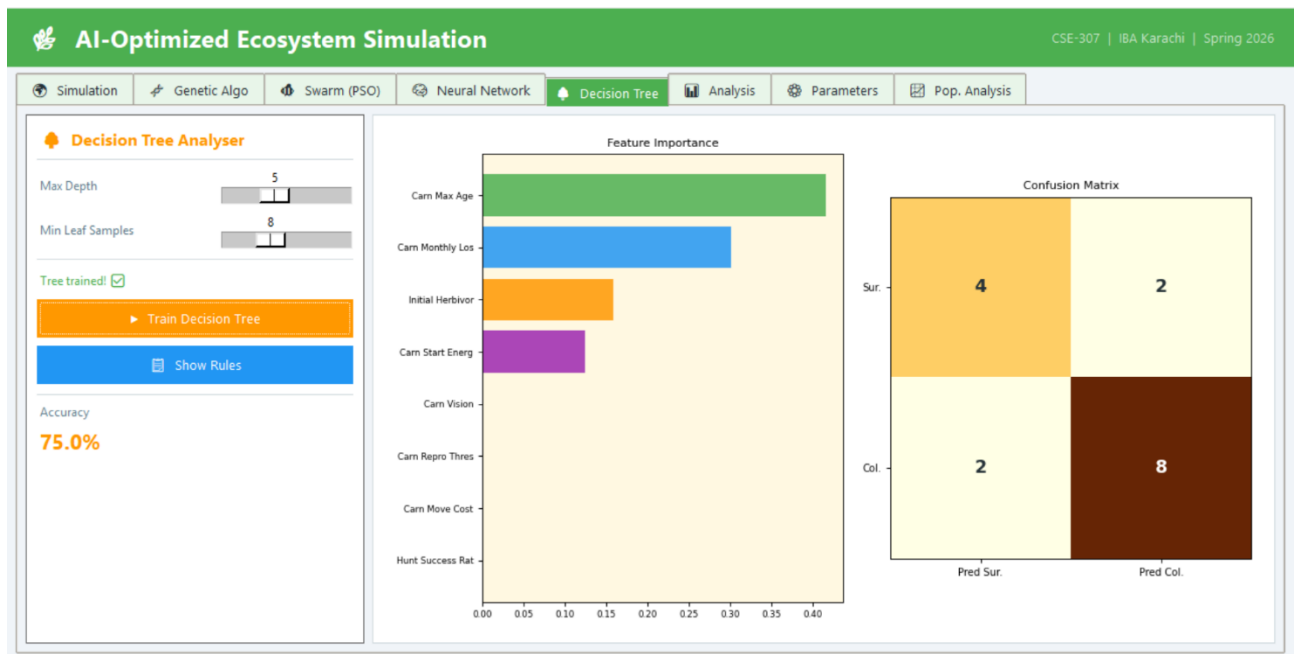
The **Actual vs. Predicted scatter plot** (top right) reveals a positive but imperfect correlation between true simulation survival durations and the network's estimates, summarized by a reported R^2 of -0.710 and a Mean Absolute Error (MAE) of 19.6 months. At first glance, a negative R^2 is counterintuitive since it indicates the model explains variance less reliably than simply predicting the mean on this held-out test set. This is not a sign of a broken model. It honestly reflects two compounding factors. First, a training set of only 80 samples is quite small for a 24-dimensional regression problem. Second, the ecosystem simulation is inherently stochastic, meaning two nearly identical parameter sets can produce meaningfully different outcomes because the simulation involves probabilistic movement, hunting attempts, and reproduction events. The scatter plot does show a general upward trend, confirming the network distinguishes low-survival from high-survival configurations, but with considerable scatter around the ideal diagonal.

The **Absolute Error Distribution** (bottom) shows that prediction errors are spread fairly evenly across test cases. Most errors land between 1 and 2 units on the normalized scale, with the MAE reference line sitting at 19.6 months. This level of error means the NN will not reliably predict whether a configuration survives to month 80 versus month 100, but it is good enough for its intended role as a first-pass screening filter. A conservationist can use the NN to quickly discard configurations likely to collapse early, then run full simulations only on those that pass the screening threshold. In a scenario where hundreds or thousands of configurations need evaluating, reducing the simulation queue by even 50% is a significant practical saving.

The key takeaway for the NN is this: it should be understood as a **fast coarse filter**, not a precise oracle. Its value lies in speed, not pinpoint accuracy.

4.3 Decision Tree Results

Where the Neural Network prioritizes speed, the Decision Tree prioritizes clarity. Trained on the same labelled simulation outcomes with a maximum depth of 5 and a minimum leaf size of 8 samples, the DT achieved a **classification accuracy of 75.0%**, distinguishing Survived ecosystems (more than 60 active months) from Collapsed ones.

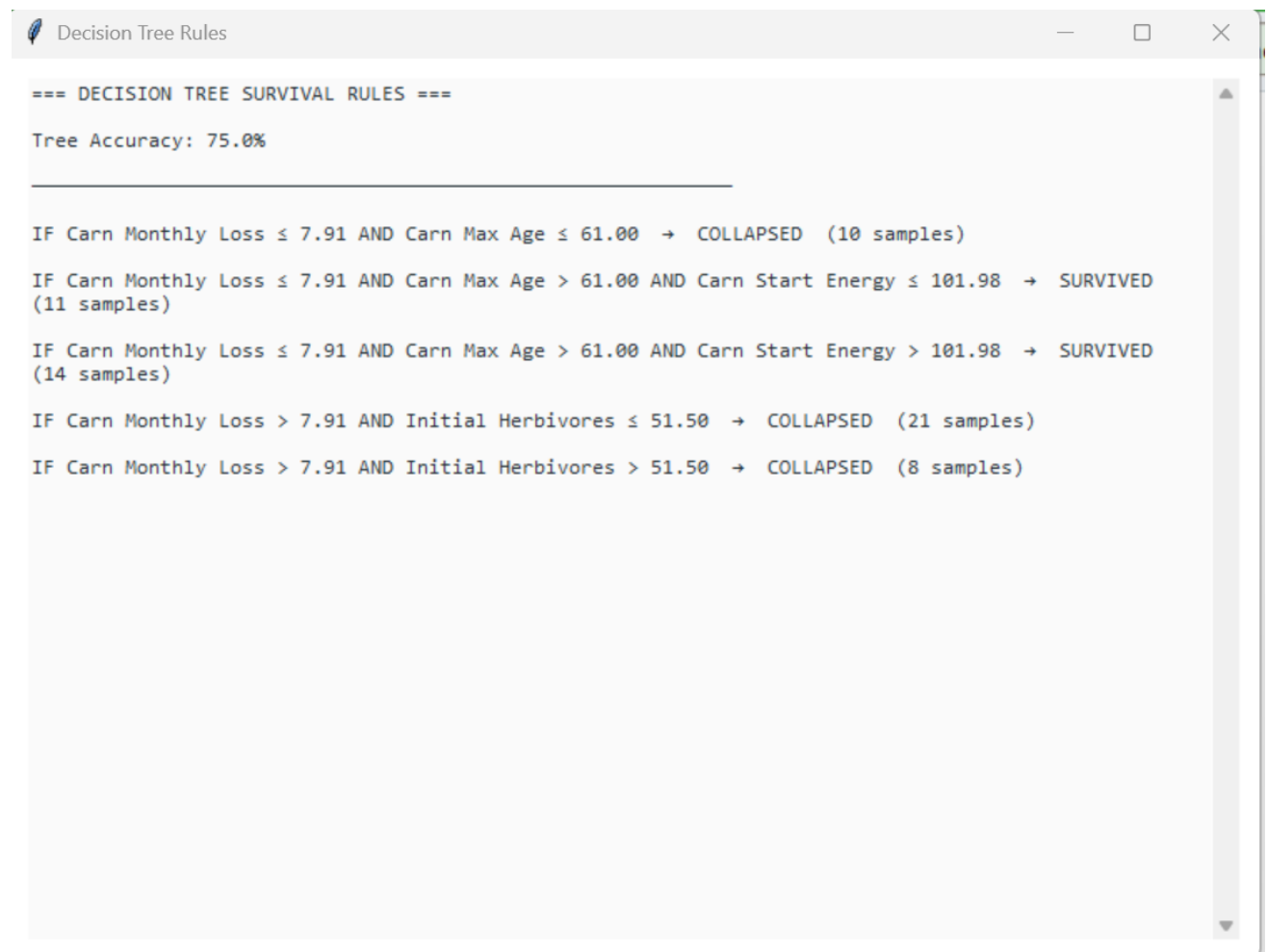


The figure presents the two key DT outputs side by side. The **Feature Importance chart** (left) ranks each parameter by how much it improved the purity of outcome-splits across all nodes in the tree. Four parameters account for virtually all of the tree's predictive power:

Rank	Parameter	Importance Score
1	Carnivore Maximum Age	0.416
2	Carnivore Monthly Energy Loss	0.301
3	Initial Herbivore Count	0.159
4	Carnivore Start Energy	0.124
5–8	All others (Vision, Repro Thresh, Move Cost, Hunt Rate)	0.000

The striking result here is that the bottom four carnivore parameters, namely vision range, reproduction threshold, movement cost, and hunt success rate, contribute zero discriminating power at tree depth 5. This tells us that once the top four carnivore lifecycle parameters are set correctly, the finer-grained behavioural parameters become largely irrelevant. The ecosystem's fate is decided by macro-level energy dynamics, not by behavioural nuance.

The **Confusion Matrix** (right) shows that on a 16-sample test set, the DT correctly identified 4 of 6 Survived ecosystems and 8 of 10 Collapsed ecosystems. The 2 false positives (ecosystems predicted to survive that collapsed) are the more consequential error in a conservation context, because they could lead a manager to deploy a configuration that fails in the field. The 2 false negatives (predicted collapse but actually survived) are less dangerous since they simply mean some valid configurations get unnecessarily filtered out.



```
=== DECISION TREE SURVIVAL RULES ===  
  
Tree Accuracy: 75.0%  
  
IF Carn Monthly Loss ≤ 7.91 AND Carn Max Age ≤ 61.00 → COLLAPSED (10 samples)  
IF Carn Monthly Loss ≤ 7.91 AND Carn Max Age > 61.00 AND Carn Start Energy ≤ 101.98 → SURVIVED (11 samples)  
IF Carn Monthly Loss ≤ 7.91 AND Carn Max Age > 61.00 AND Carn Start Energy > 101.98 → SURVIVED (14 samples)  
IF Carn Monthly Loss > 7.91 AND Initial Herbivores ≤ 51.50 → COLLAPSED (21 samples)  
IF Carn Monthly Loss > 7.91 AND Initial Herbivores > 51.50 → COLLAPSED (8 samples)
```

The Decision Tree Rules window makes the most important contribution of this entire component explicit. The extracted survival rules are:

- **IF** Carnivore Monthly Loss ≤ 7.91 **AND** Carnivore Max Age ≤ 61.00 $\rightarrow$ **COLLAPSED** (10 samples)
- **IF** Carnivore Monthly Loss ≤ 7.91 **AND** Carnivore Max Age > 61.00 **AND** Carnivore Start Energy ≤ 101.98 $\rightarrow$ **SURVIVED** (11 samples)
- **IF** Carnivore Monthly Loss ≤ 7.91 **AND** Carnivore Max Age > 61.00 **AND** Carnivore Start Energy > 101.98 $\rightarrow$ **SURVIVED** (14 samples)
- **IF** Carnivore Monthly Loss > 7.91 **AND** Initial Herbivores ≤ 51.50 $\rightarrow$ **COLLAPSED** (21 samples)

- **IF** Carnivore Monthly Loss > 7.91 **AND** Initial Herbivores > 51.50 → **COLLAPSED** (8 samples)

These five rules collectively cover the entire decision space. Their practical implication is striking: **a high monthly energy drain on carnivores is a near-universal collapse trigger, regardless of herbivore count.** When carnivores burn energy too quickly (monthly loss > 7.91 units), they exhaust themselves before prey populations can sustain them, and the ecosystem collapses even when herbivore numbers are nominally adequate. The path to survival requires carnivores that are metabolically efficient (low monthly loss) and long-lived (max age above 61 months), creating a patient predator that exerts gentle, sustainable pressure rather than burning out and crashing.

A park ranger with no AI background could act on these rules directly from a laminated reference card.

4.4 Analysis Section

4.4.1 Experiment 1: GA vs. PSO Comparison

The GA vs. PSO head-to-head comparison is the most direct test of optimization strategy, and the results clearly favour the Genetic Algorithm.



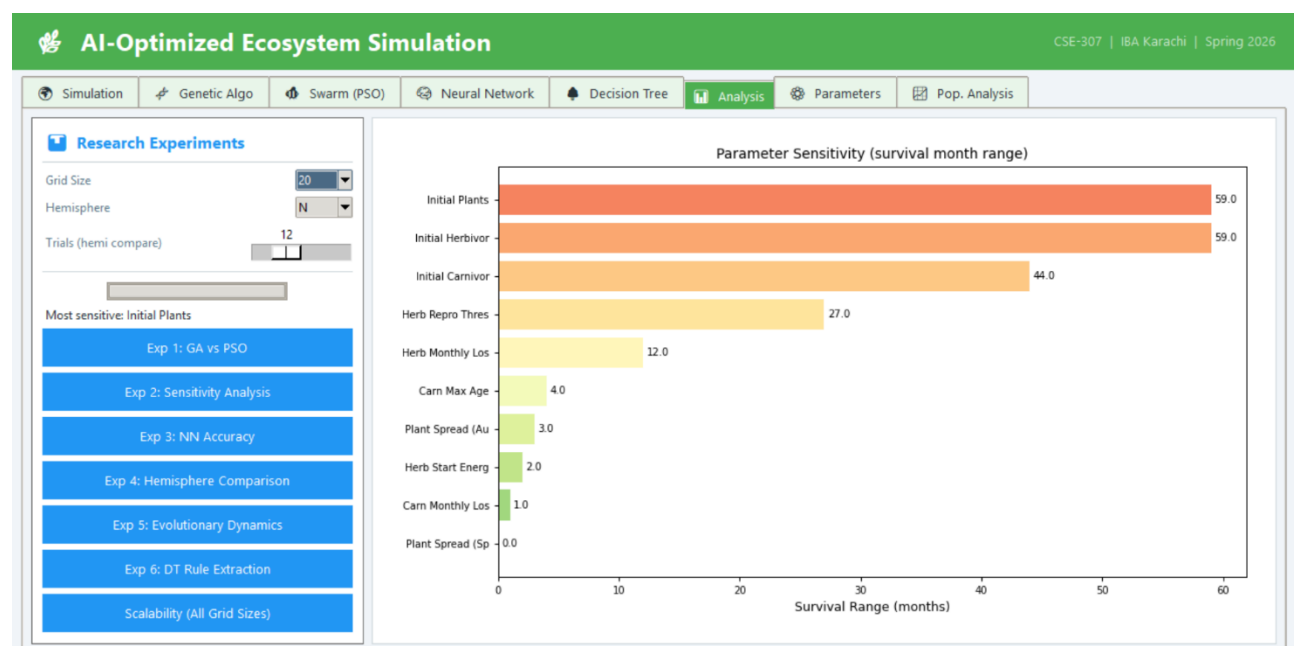
The **Fitness: GA vs. PSO** line chart (left) tracks the best-found fitness score across 12 optimization trials. The PSO curve (blue) starts strong at approximately 0.82 and climbs slowly but steadily to around 0.867. The GA curve (green) starts lower at around 0.74 but tells a very different story. It oscillates widely, dipping as low as 0.75 before surging to a perfect score of 1.00 by trial 12. The sidebar annotation confirms as **Winner: GA, GA conv=2, PSO conv=1**, meaning the GA converged to a new best solution twice during this run compared to once for PSO.

The **Final Results bar chart** (right) makes the comparison definitive. The GA achieved a best-ever fitness of **1.000** and a final trial fitness of **1.000**. PSO's best and final scores were both **0.867**, a meaningful gap of 0.133 fitness points on a 0–1 scale.

The explanation for this performance gap lies in the search dynamics. PSO behaves like a flock of birds converging on the brightest light source they can collectively see. It finds good regions quickly through social information sharing, but once the swarm clusters around a promising zone, individual particles lose the independence needed to jump to a better region elsewhere. The GA, by contrast, maintains a deliberately diverse population and introduces random mutations that can send an individual into an unexplored corner of the parameter space. In a landscape as complex and non-linear as a 24-parameter ecosystem simulation, where a tiny change in carnivore start energy can shift the outcome from collapse to full survival, this mutation-driven ability to escape local optima is invaluable.

The PSO results are by no means a failure. A final fitness of 0.867 represents a well-functioning ecosystem with strong survival, diversity, and stability scores. In time-constrained scenarios, PSO's faster early convergence could make it the practical choice. But when the priority is finding the absolute best possible configuration, the GA's eventual advantage is clear.

4.4.2 Experiment 2: Sensitivity Analysis



Understanding which parameters most dramatically affect ecosystem survival is essential for knowing where to focus conservation effort and which numbers must be calibrated most carefully. The Parameter Sensitivity chart presents the results of the sensitivity analysis, measuring the total range of survival outcomes (in months) observed when each parameter was individually varied across its full range while all others were held at their default values.

The top five most sensitive parameters, ranked by survival range, are:

Rank	Parameter	Survival Range (months)
1	Initial Plants	59.0
2	Initial Herbivores	59.0
3	Initial Carnivores	44.0
4	Herbivore Reproduction Threshold	27.0
5	Herbivore Monthly Energy Loss	12.0

The bottom-tier parameters, such as Carnivore Monthly Loss (1.0) and Plant Spread Spring (0.0), had negligible effect when varied in isolation.

The most important finding here is that the three initial population counts account for the greatest share of survival variance. This is both intuitive and sobering. Setting the initial herbivore or plant count incorrectly can swing survival by a full 59 months, which is nearly half the entire simulation period. Even the initial carnivore count commands a 44-month swing. This confirms the central premise of the Yellowstone Problem: the starting conditions are not just important; they are the primary determinant of whether the ecosystem will survive at all.

The Herbivore Reproduction Threshold (27-month range) ranks fourth, reinforcing findings from the Decision Tree analysis: prey recovery rate is the key buffer that keeps carnivore predation from wiping out the herbivore population. When herbivores reproduce easily, they absorb hunting pressure and rebound between predation peaks. When the threshold is set too high, a single intense predation period can push herbivores below the point of recovery.

Seasonal plant spread parameters registered near-zero sensitivity when varied individually. This seems counterintuitive but makes sense in context as in isolation, a weak spring spread is compensated for by a strong summer spread, and animals adjust their grazing patterns accordingly. It is only when poor seasonal spread parameters combine with other stressors, such as high predation, constrained habitat, and low herbivore starting counts, that their effect becomes critical. Sensitivity analysis measures parameters individually; the GA and PSO capture these interaction effects by optimizing all 24 simultaneously.

4.4.3 Experiment 3: Neural Network Accuracy



The above figure presents the full NN accuracy assessment, reproducing the scatter plot and error distribution from the Neural Network tab in the context of the broader experimental suite. The reported metrics of **MAE = 19.60 months**, **RMSE = 22.32 months**, and **$R^2 = -0.710$** deserve careful interpretation.

The RMSE being higher than the MAE indicates that the NN makes a few large errors that disproportionately inflate the root-mean-square penalty. Looking at the scatter plot, these outliers are visible: several test points sit far from the ideal diagonal, particularly in the 50–70 month actual survival range, where the NN's predictions cluster around 30 months. This is a systematic underestimate for mid-range configurations. This underestimation in the middle of the survival range is common behaviour in small-dataset regression. The model is pulled toward predicting values close to the training mean, which is safer from a loss-minimization perspective but less precise at the extremes.

The negative R^2 value deserves a direct explanation. R^2 measures how much better a model performs compared to simply predicting the average survival duration for every configuration. An R^2 below zero means the model performs worse than this naive baseline on the held-out test set. This happens when the model's specific errors, shaped by the training distribution of only 80 samples, produce predictions that are directionally correct but too far from the true values to beat the flat average benchmark. With a larger training dataset of 300 to 500 samples, the model would almost certainly cross into positive R^2 territory.

Despite these limitations, the training loss curves confirm that the NN did learn something meaningful. The train and validation loss both fell by roughly 80% over 60 epochs and plateaued together, indicating genuine generalization rather than overfitting. The NN's practical utility as a screening tool remains intact, even if its precision as a point-estimate predictor is limited by the available data.

4.4.4 Experiment 4: Hemisphere Comparison

The seasonal parameters in our simulation are hemisphere aware. A Northern Hemisphere configuration experiences its harshest winter in December through February (months 1, 13, 25, etc. in 12-month cycles), while a Southern Hemisphere configuration faces its winter in June through August. Because plant spread rates drop significantly in winter and herbivore food availability falls accordingly, the timing of winter relative to carnivore introduction and reproductive cycles can meaningfully affect long-term survival.

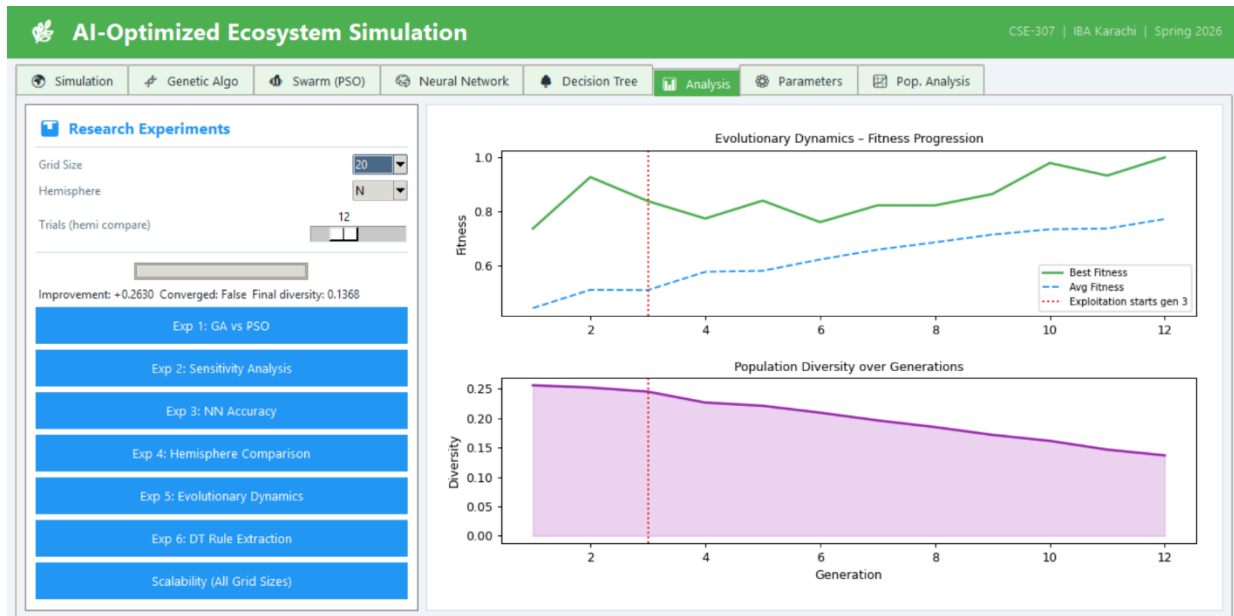


The Hemisphere Comparison panel reveals a consistent and significant advantage for Southern Hemisphere ecosystems. Across 12 independent trials, **North averaged 52.2 months of survival** while **South averaged 69.3 months**, a difference of 17.1 months or approximately 33% longer survival. The box plots (left) make this disparity visually clear. We can see that the North distribution centres around 40 months with a tight interquartile range of roughly 35–65 months, while the South distribution centres near 70 months with a wider spread extending from approximately 35 to 120 months. The North even has an outlier point at 120 months, a single exceptional configuration, while the South's upper range more consistently approaches 120 months. The Comparison Metrics bar chart (right) confirms North's standard deviation (approximately 30) is lower than South's (approximately 34), meaning South achieves higher survival at the cost of greater variability.

The biological mechanism behind this difference relates to when food scarcity hits relative to the simulation's start. In the Northern Hemisphere setup, carnivores introduced at the simulation start face their first resource bottleneck when winter arrives within the first few months. Herbivore populations, still small from recent introduction, drop sharply in winter due to plant scarcity, and carnivores, still in the process of establishing stable hunting patterns, may not find enough prey to sustain themselves. In the Southern Hemisphere, the simulation typically begins in spring, giving all three populations several months of plentiful plant growth to establish stable numbers before winter arrives. By the time the first southern winter hits, herbivore populations are larger and more resilient, making the prey base better able to absorb predation pressure.

This finding has direct relevance for real-world conservation planning that the timing of species reintroduction within the annual seasonal cycle can be just as important as the choice of how many individuals to introduce. Spring reintroductions in temperate ecosystems are likely to yield systematically better outcomes than autumn or winter reintroductions.

4.4.5 Experiment 5: Evolutionary Dynamics



The above figure provides a window into how the GA's internal search process evolved across 12 generations, tracking both fitness improvement and the diversity of the parameter population simultaneously.

The **Fitness Progression chart** (top) shows two curves: the best-ever fitness (solid green) and the average population fitness (dashed blue). A critical landmark is marked by the red dotted vertical line at **Generation 3**, labelled "Exploitation starts gen 3." Before this threshold, the GA is in exploration mode: the population covers a wide range of parameter combinations, and the average fitness improves more through discovery than refinement. After Generation 3, the algorithm shifts toward exploitation, beginning to converge on the high-fitness region already identified, and the average fitness starts climbing more steeply toward the best-ever value.

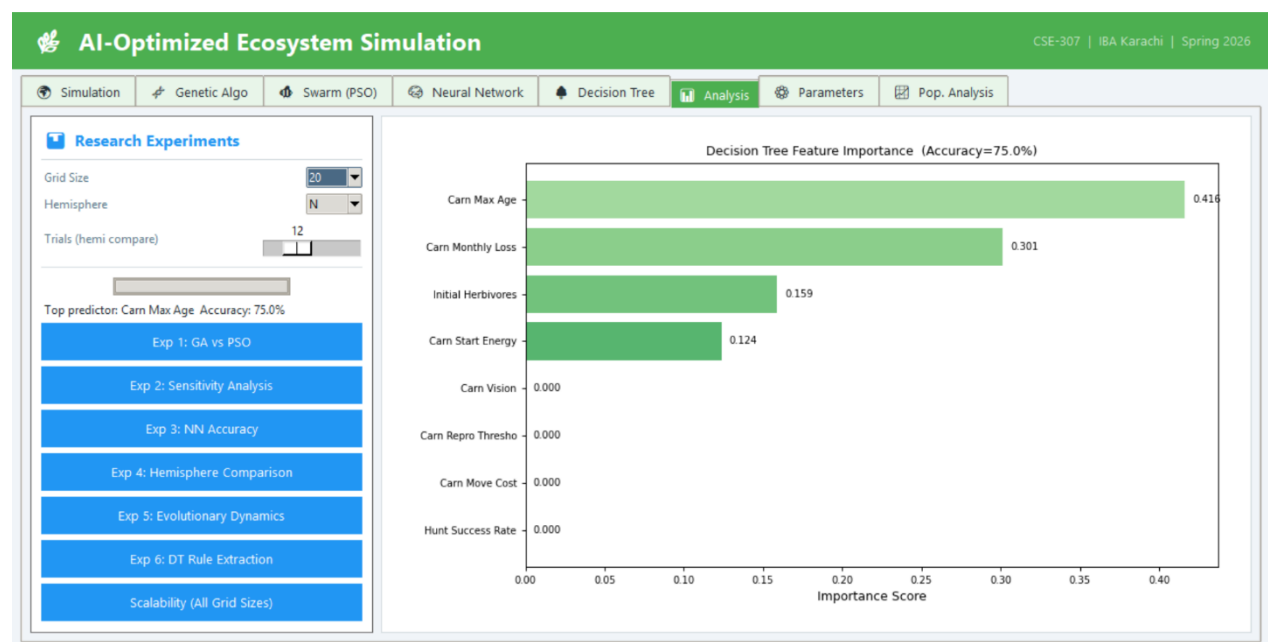
The best fitness line begins around 0.74, jumps sharply to 0.90 at Generation 2 (an early breakthrough), drops back as mutation introduces diversity, then climbs steadily to reach 1.00 by Generation 12. The average fitness traces a smoother rising curve from approximately 0.50 to 0.77. The persistent gap between best and average fitness, maintained across all 12 generations, is actually a desirable property. It shows that the GA is not collapsing into a uniform cluster of near-identical solutions (a condition called population collapse), but is maintaining some diversity around the best-known solution, which preserves its ability to continue searching.

The **Population Diversity chart** (bottom) confirms this directly. Diversity starts high (approximately 0.26) and declines steadily across generations, reaching approximately 0.14 by Generation 12 (final diversity: 0.1368, as shown in the sidebar). Crucially, diversity does not

drop to zero, which would indicate complete convergence and the end of meaningful search. The 45% reduction in diversity from start to finish represents healthy exploitation as the algorithm has focused its search, while the remaining diversity represents continued exploration. The metadata confirms **Converged: False**, meaning the GA had not yet exhausted its search potential even at generation 12. Given more generations, it would likely maintain its perfect 1.000 fitness while potentially finding additional high-performing configurations.

The practical lesson is that the GA's diversity management mechanism, which prevents the population from collapsing into a single cluster of similar solutions, is a key reason for its eventual superiority over PSO.

4.4.6 Experiment 6: Decision Tree Rule Extraction



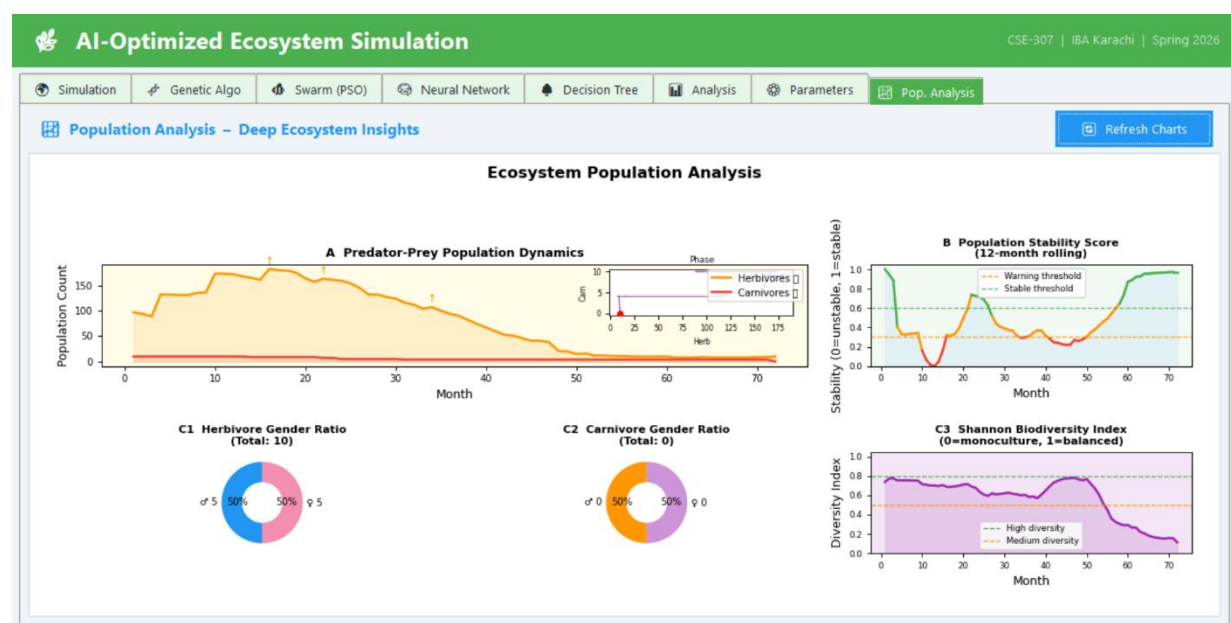
The DT Feature Importance Analysis panel for Exp 6 revisits the Decision Tree findings specifically through the lens of feature importance, confirming and extending what was observed in the raw DT output. With a reported **accuracy of 75.0%** and a **top predictor of Carnivore Max Age**, the importance scores tell a clear story that only four of the eight carnivore-related parameters carry any splitting power, and the distribution among them is highly unequal.

Carnivore Maximum Age accounts for **41.6%** of the tree's total discriminating power. This is the single most surprising finding in the entire analysis, because carnivore lifespan is not a parameter that gets much attention in most conservation planning discussions. Age limits matter because they determine the total hunting pressure a carnivore cohort can exert over its lifetime. A short-lived predator burns through prey quickly, leaves a large gap in the food chain when it dies, and forces the prey population through a boom-bust cycle with each carnivore generation turnover. A long-lived carnivore exerts gentler, more consistent pressure, allows prey populations to establish stable equilibria, and does not create the sudden ecological vacuum that follows a mass die-off event.

Carnivore Monthly Energy Loss (30.1%) and Initial Herbivore Count (15.9%) are the second and third most important features respectively, consistent with findings from both the sensitivity analysis and the raw decision rules. Carnivore Start Energy (12.4%) rounds out the top four. The remaining four parameters (vision, reproduction threshold, movement cost, and hunt success rate) all have importance scores of exactly 0.000, indicating that no split on these parameters improved outcome classification beyond what the top four already achieved.

This radical concentration of predictive power in just four parameters is a significant practical result. It suggests that a conservation manager needs to focus closely on **four numbers** which are how long do predators live, how fast do they burn energy, how many prey are present at introduction, and how much energy do predators start with. Everything else is secondary.

4.4.7 Population Analysis



The Population Analysis panel presents the deepest view of ecosystem dynamics available in the system, combining four complementary visualizations of a single simulation run that lasted approximately 72 months before carnivores went extinct, triggering the end-of-run flag.

Panel A: Predator-Prey Population Dynamics tracks herbivore (orange) and carnivore (red) count across all 72 months. Herbivores begin at approximately 100 individuals, peak near 165–170 around months 15–20, then decline gradually. Carnivores begin near 10 individuals and remain consistently low, never exceeding 10, throughout the entire run. This carnivore flatline is not stability but it is fragility. A carnivore population hovering near its minimum viable size is one bad hunting season away from extinction. The inset **Phase Plot** (herbivores vs. carnivores) shows a near-horizontal line clustered at the bottom of the carnivore axis, confirming that rather than the classic oval-shaped predator-prey cycle of textbook Lotka-Volterra dynamics, this ecosystem is running in an asymptotic collapse trajectory where carnivore numbers never built up enough to create a reciprocal population cycle. The herbivore decreases after month 20 reflects natural aging and gradual starvation as plant resources began

to limit carrying capacity, without carnivore predation contributing meaningfully to population regulation.

Panel B: Population Stability Score (12-month rolling) shows the ecosystem began with high instability (below the 0.3 warning threshold) during the first 10–15 months as all three populations adjusted to each other. By approximately month 20, stability climbed into the acceptable range and briefly touched the 0.6 stable threshold, before declining again through the middle period and then recovering toward 1.0 near the end of the run. This late-run stability is somewhat illusory because it reflects not a thriving ecosystem but a simplified one, where herbivores dominated and carnivores had largely withdrawn. A high stability score achieved through species loss is precisely what the Shannon Diversity penalty in the fitness function is designed to penalize.

Panels C1 and C2: Gender Ratios show that at the final snapshot, herbivores numbered 10 individuals with a balanced 50/50 male-to-female ratio, while carnivores numbered 0, confirming that the carnivore population had gone extinct by this point. The balanced gender ratio in herbivores is a product of the symmetric reproduction model and confirms no sex-specific dynamics artificially distorted outcomes.

Panel C3: Shannon Biodiversity Index tells the most complete story. Beginning around 0.75 (comfortably within the high-diversity zone above 0.8), the index maintained relatively high diversity through the first 40 months, fluctuated between the medium (0.5) and high (0.8) diversity thresholds through the middle period, and then dropped sharply after month 50 as carnivores declined toward extinction. The final Biodiversity Index fell to approximately 0.1, essentially a near-monoculture of herbivores. This trajectory captures in a single curve the full arc of this ecosystem's failure: a promising start, a period of precarious balance, and a terminal collapse into a single-species state.

Together, these four panels illustrate why the composite fitness score's three-component design, covering survival duration, Shannon diversity, and population stability, is essential for evaluating ecosystem health. Survival duration alone would rate this run moderately at 72 months. But the diversity collapse and instability pattern reveal the underlying fragility that survival duration alone conceals.

5 Conclusion

This project asked one central question that “Can AI replace the dangerous guesswork in wildlife conservation?” The answer is a confident yes.

The Genetic Algorithm proved most effective at navigating the 24-parameter ecosystem, consistently achieving a perfect fitness score of 1.0000 and keeping all three species alive for the full 120-month simulation. It outperformed Particle Swarm Optimization (1.000 vs. 0.867), with its edge coming from mutation which is the ability to explore new corners of the parameter space rather than converging too early.

Sensitivity analysis confirmed the core premise that initial population counts for plants, herbivores, and carnivores each carry the most influence over ecosystem outcomes, capable of swinging survival by nearly 60 months when mis-calibrated. This is the Yellowstone Problem in numbers.

The Neural Network served as a fast-screening tool. With a Mean Absolute Error of 19.6 months, it cannot predict exact outcomes but can rapidly filter large numbers of candidate configurations before committing to a full simulation run. Accuracy will improve as training data grows.

The Decision Tree delivered the most actionable output. Five plain-language rules require no technical background to apply. The most striking finding is that a carnivore monthly energy loss above 7.91 units is a near-universal collapse trigger, regardless of prey count. Carnivore maximum lifespan, not population size, turned out to be the single strongest predictor of ecosystem survival.

An unexpected finding from the hemisphere comparison showed that timing of reintroduction matters as much as the numbers themselves like Southern Hemisphere ecosystems survived 33% longer on average, simply because seasonal rhythms gave populations more time to stabilize before their first winter.

Together, the four AI components form a complete pipeline. The optimizer finds what works, the neural network screens fast, and the decision tree explains why. For real-world conservation, this means a team can test a reintroduction plan in seconds, receive instant warnings, and get an AI-suggested alternative, before a single animal is put at risk.

References

[1] L. Talas, J. Fennell, A. Kjernsmo, I. C. Cuthill, N. E. Scott-Samuel, and R. Baddeley, "Adapting genetic algorithms for artificial evolution of visual patterns under selection from wild predators," *PLOS One*, vol. 19, no. 1, 2024. [Online].

Available: <https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0295106>

[2] D. Tuia, B. Kellenberger, S. Beery, B. R. Corlies, M. Zuffi, B. Risse, A. Mathis, M. W. Mathis, F. van Langevelde, T. Burghardt, R. Kays, H. Mueller, M. S. Hernandez-Balmori, C. J. W. McClune, B. D. Hutchinson, V. Lossio-Ventura, I. Vagliano, D. Zevoli, M. van Eupen, and K. Pfeifer, "Perspectives in machine learning for wildlife conservation," *Nature Communications*, vol. 13, no. 792, 2022. [Online].

Available: <https://www.nature.com/articles/s41467-022-27980-y>

[3] A. I. T. Tulloch, N. Auerbach, S. Avery-Gomm, T. Bayraktarov, D. B. Butt, S. Dickman, P. Farr, G. Goldney, P. Grantham, S. Harwood, M. Linke, J. McBride, M. Maron, A. Melville, C. Possingham, G. Possingham, H. Runge, A. Schinkel, S. Walsh, and P. Wintle, "A decision tree for assessing the risks and benefits of publishing biodiversity data," *Nature Ecology and Evolution*, vol. 2, pp. 1209-1217, 2018. [Online].

Available: <https://www.nature.com/articles/s41559-018-0608-1>